

MONTREAL TRUDEAU, QC, Canada

WMO: 716270

Lat: 45.4667N

Lon: 73.7500W

Elev: 36

StdP: 100.89

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-20.2	-29.5	0.2	-22.1	-26.7	0.3	-19.8	13.3	-3.7	12.0	-6.0	4.3	260	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	30.3	22.1	28.7	21.1	27.4	20.4	23.4	28.2	22.5	26.8	21.6	25.7	5.4	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.8	16.6	26.1	21.0	15.7	25.0	20.1	14.8	24.5	70.0	28.1	66.3	26.9	63.2	25.7	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.3	9.9	8.8	-25.7	32.8	2.8	1.5	-27.7	33.8	-29.4	34.7	-31.0	35.5	-33.1	36.6		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-26.0	25.5	2.7	1.0	-28.0	26.2	-29.6	26.8	-31.1	27.3	-33.1	28.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-8.8	-7.2	-1.5	6.4	14.1	19.2	21.8	20.8	16.7	9.4	2.5	-4.4
	DBStd	11.77	6.98	6.00	5.72	4.65	4.26	3.65	2.79	2.97	4.04	4.44	4.93	5.99
	HDD10.0	2303	584	481	359	127	11	0	0	0	2	66	228	447
	HDD18.3	4257	842	714	616	359	144	32	4	11	78	277	475	704
	CDD10.0	1387	0	0	1	18	137	277	366	336	201	48	3	0
	CDD18.3	301	0	0	0	1	12	59	112	88	28	1	0	0
	CDH23.3	2001	0	0	1	7	106	417	759	536	171	4	0	0
	CDH26.7	450	0	0	0	1	20	100	183	111	35	0	0	0
Wind	WSAvg	4.2	4.7	4.6	4.5	4.7	4.3	3.9	3.8	3.6	3.7	4.2	4.4	4.5
Precipitation	PrecAvg	987	76	65	72	80	73	84	88	94	90	79	95	87
	PrecMax	1247	151	178	115	177	140	144	186	173	232	183	195	167
	PrecMin	825	24	11	19	14	16	22	30	24	24	17	32	26
	PrecStd	109	32	33	26	36	30	30	38	38	43	36	36	33
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	7.2	15.6	24.2	29.0	31.6	32.3	31.4	30.3	23.8	16.7	10.7
		MCWB	7.3	4.9	9.8	15.7	19.0	22.7	23.5	22.9	22.4	17.5	13.7	9.3
	2%	DB	4.9	4.7	10.7	19.7	26.2	29.4	30.3	29.2	27.3	20.8	14.0	7.3
		MCWB	3.5	2.8	6.5	11.9	17.3	21.1	22.4	21.5	20.5	16.2	11.5	5.8
	5%	DB	3.0	3.1	7.9	16.6	23.9	27.5	28.7	27.8	25.2	18.2	11.6	5.1
		MCWB	1.8	1.6	4.4	10.2	15.7	20.0	21.5	20.7	19.5	14.1	9.1	3.6
	10%	DB	1.1	1.6	5.9	13.8	21.5	25.8	27.2	26.3	23.3	16.1	9.4	3.2
		MCWB	0.0	0.3	3.1	8.5	14.8	19.0	20.6	20.1	18.6	12.9	7.1	1.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	5.6	11.7	16.8	21.0	24.1	25.1	24.1	23.3	19.2	14.5	9.4
		MCDB	7.8	7.1	14.6	22.5	26.8	29.5	30.8	29.8	27.9	22.5	15.9	10.6
	2%	WB	3.9	3.2	7.2	13.3	19.1	22.5	23.7	22.9	21.6	16.6	11.8	6.0
		MCDB	4.7	4.4	10.1	17.8	23.9	27.4	28.3	27.4	25.7	19.8	13.8	7.2
	5%	WB	1.9	1.7	5.0	11.0	17.5	21.3	22.6	22.0	20.4	14.8	9.4	3.8
		MCDB	2.9	2.9	7.6	15.7	21.8	25.8	26.9	26.0	24.3	17.7	11.3	4.9
	10%	WB	0.1	0.4	3.3	9.1	15.8	20.1	21.7	21.1	19.2	13.1	7.3	2.0
		MCDB	1.2	1.6	5.5	13.2	19.8	24.4	25.8	24.8	22.8	15.6	9.2	3.0
Mean Daily Temperature Range	5% DB	MDBR	7.6	7.9	7.9	9.2	9.9	9.2	8.8	8.9	9.1	7.7	6.8	6.5
		MCDBR	8.8	8.4	10.3	14.1	12.9	11.2	10.1	10.0	10.8	10.7	10.0	7.5
		MCWBR	8.7	7.7	7.5	8.7	6.8	5.7	5.2	5.0	5.9	7.0	8.3	7.2
	5% WB	MCDBR	8.6	8.1	9.8	12.7	10.7	9.6	8.8	8.8	9.7	9.6	9.6	7.4
		MCWBR	8.6	7.7	7.8	8.8	6.8	5.9	5.3	5.0	6.0	7.1	8.5	7.2
Clear-Sky Solar Irradiance	taub	0.261	0.289	0.332	0.365	0.390	0.446	0.431	0.416	0.375	0.338	0.308	0.292	
	taud	2.344	2.257	2.290	2.358	2.348	2.196	2.246	2.284	2.385	2.494	2.489	2.308	
	Ebn at Noon	858	889	895	895	879	827	834	833	841	831	798	764	
	Edh at Noon	75	100	113	116	121	141	133	123	103	80	67	71	
All-Sky Solar Radiation	RadAvg	1.44	2.39	3.64	4.39	5.13	5.63	5.67	5.02	3.97	2.30	1.40	1.00	
	RadStd	0.16	0.24	0.30	0.50	0.49	0.43	0.43	0.23	0.33	0.20	0.17	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page